Jaden Crocker

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Education

Washington University in St. Louis — St. Louis, MO
M.S. Aerospace Engineering, ***GPA: 4.0*** — *Expected – May 2027*
B.S. Mechanical Engineering, ***GPA: 3.8*** | *Boeing Engineering Scholar* — *May 2026*

Creighton University — Omaha, NE
B.S. Physics; minor in Mathematics, ***GPA: 3.83/4.0*** — *May 2026*

Notable Coursework: Aircraft Performance, Aircraft Design, Computational Fluid Dynamics, Fluid Dynamics, Fatigue and Fracture Analysis, Mechanical Behavior of Composites

Engineering Experience

FSAE WashU Racing — *Washington University in St. Louis*
Chief Engineer — *May 2026 – Present*

- Set vehicle-level performance goals each season by analyzing prior competition results and vehicle data, identifying suspension and powertrain inconsistencies as this season's primary drivers of performance degradation
- Manage the technical design, build, integration, and testing of the vehicle's six technical systems across a **60+ member team**, hosting thorough subsystem Final Design Reviews that gate release to manufacturing
- Plan, organize, and run the vehicle test program, including test plan & procedure creation and driver selection & training, raising usable-data rate per test day to **70%**, up from 50%
- Developing the team's first ever **5-Dimensional Aerodynamic Map** using an HPC-based Computational Fluid Dynamics (CFD) workflow that **cut simulation time by >75%**
- Led the development and integration of a real-time fuel consumption calculator aimed at previous years' highest-risk failure mode, **verified within 4% error** using on-track consumption data

Aerodynamics and Composites Lead — *May 2025 – May 2026*

- Set vehicle-level aerodynamic targets (C_L **= -3.2**, C_D **= 1.3**, CoP **= 40-45% Front**) using OptimumG lap-time simulation, translating full-vehicle performance goals into system-level engineering requirements
- Defended the Aerodynamics system's engineering, trade-off, and design decisions to industry judges at the Formula SAE Design Event, scoring a system-high **13/15 points**
- Validated ANSYS Fluent CFD simulation data (C_L, C_D, CoP, and pitch/roll sensitivity envelopes) against wind-tunnel and on-track data to **within 10%**, closing the loop between simulation and real-world vehicle performance
- Led the composite manufacturing of **16 system components** through a wet layup carbon fiber lamination process
- Led a full redesign of the aerodynamic package's mounting system and cockpit body panels, cutting mass **∼7%** below target, reducing mean service time, and improving technician access to critical areas

Bergstrom Inc. — *Rockford, IL*
CFD Engineer Intern — *May 2026 – August 2026*

- Built an automated CFD optimization workflow using ANSYS Discovery and ANSYS OptiSLang, cutting design iteration lead time from **1-2 weeks to 3-4 hours**
- Applied the workflow to a blower-fan HVAC unit and a 10-outlet air distribution duct, improving evaporator face-velocity uniformity from **65% to 91%** and cutting maximum flow imbalance from **52% to 6%**

Washington University in St. Louis — *St. Louis, MO*
Automated EV Charger Delivery Arm (Senior Design Project) — *August 2025 – December 2025*

- Designed and manufactured a lead screw driven, five-link scissor lift mechanism with a **10:1 extension-to-retraction ratio** for a local autonomous EV charging start-up
- Conducted frictional torque analysis to specify a stepper motor and ball screw system selection, ensuring a **1.0 N·m torque safety factor** with 5 lb end-effector payload

Professional Experience

Crocker's Collision Center — *Watertown, SD*
Autobody Technician — *May 2013 – August 2023, Seasonal*

- Diagnosed complex structural failure modes in collision-damaged vehicles, executing start-to-finish repairs involving high-precision fabrication, bodywork, welding, and refinishing

Technical Skills

Design & Modeling: SOLIDWORKS, Autodesk Fusion 360, Rhino3D, ANSYS DesignModeler, ANSYS SpaceClaim, ANSYS Fluent Meshing
Simulation & Optimization: ANSYS Fluent (CFD), ANSYS OptiSLang, ANSYS Ensight, Simulink, ANSYS Discovery, Lap Time Simulation (OptimumG)
Programming & Computation: MATLAB, HPC cluster computing, Python, LaTeX
Fabrication: Welding (TIG/MIG), 3D Printing, Precision Metal Fabrication, Composite Manufacturing, Mold Design